

Bootstrap 5 Hands-On

1. Setting Up Bootstrap 5

Exercise 1.1

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Bootstrap CDN Setup</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
rel="stylesheet">
</head>
<body>
  <div class="container mt-5">
    <h1>Hello Bootstrap 5</h1>
    <button class="btn btn-primary">Click Me</button>
  </div>
  <script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle.min.js"></script>
</body>
</html>
```

Exercise 1.2

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Bootstrap Local Setup</title>
  <link href="css/bootstrap.min.css" rel="stylesheet">
</head>
<body>
  <div class="container mt-5">
    <h1>Bootstrap loaded from local files</h1>
    <button class="btn btn-success">Local Bootstrap Button</button>
  </div>
  <script src="js/bootstrap.bundle.min.js"></script>
</body>
</html>
```

2. Bootstrap Structure and Files

Exercise 2.1

css folder - contains bootstrap.css, bootstrap.min.css and grid/utility stylesheets.

js folder - contains bootstrap.js, bootstrap.bundle.min.js (includes Popper) for components like modal, dropdown, carousel.

icons folder - contains the Bootstrap Icons svg/font files used for icon classes.

Exercise 2.2

```
<link href="css/bootstrap.min.css" rel="stylesheet">
...
<script src="js/bootstrap.bundle.min.js"></script>
```

3. Fundamentals of Responsive Grid Layout

Exercise 3.1

```
<div class="container">
  <div class="row">
    <div class="col-12 col-sm-6 col-md-4">Column 1</div>
    <div class="col-12 col-sm-6 col-md-4">Column 2</div>
    <div class="col-12 col-sm-6 col-md-4">Column 3</div>
  </div>
</div>
```

Exercise 3.2

```
<div class="container">
  <div class="row">
    <div class="col-md-6 col-lg-4">Box A</div>
    <div class="col-md-6 col-lg-4">Box B</div>
    <div class="col-md-12 col-lg-4">Box C</div>
  </div>
</div>
```

4. Column Layouts and Grid Classes

Exercise 4.1

```
<div class="container">
  <div class="row">
    <div class="col-md-3 bg-light">Sidebar</div>
    <div class="col-md-9">Content Area</div>
  </div>
</div>
```

Exercise 4.2

```
<div class="container">
  <div class="row">
    <div class="col-sm-3">Column 1</div>
    <div class="col-sm-3">Column 2</div>
    <div class="col-sm-3">Column 3</div>
    <div class="col-sm-3">Column 4</div>
  </div>
</div>
```

5. Alignment and Reordering in Grid

Exercise 5.1

```
<div class="container">
  <div class="row justify-content-center align-items-center" style="height:200px;">
    <div class="col-md-4 text-center">Centered Content</div>
  </div>
</div>
```

Exercise 5.2

```
<div class="container">
  <div class="row">
    <div class="col-md-6 order-md-2">Second on desktop</div>
    <div class="col-md-6 order-md-1">First on desktop</div>
  </div>
</div>
```

6. Responsive Flexbox Utilities

Exercise 7.1

```
<nav class="d-flex flex-column flex-md-row bg-dark p-3">
  <a class="text-white me-md-3" href="#">Home</a>
  <a class="text-white me-md-3" href="#">About</a>
  <a class="text-white me-md-3" href="#">Contact</a>
</nav>
```

Exercise 7.2

```
<div class="card">
  <div class="card-body d-flex justify-content-between align-items-center">
    <span>Card Title</span>
    <button class="btn btn-sm btn-primary">Action</button>
  </div>
</div>
```

7. Typography

Exercise 7.1

```
<h1 class="display-1">Display 1 Heading</h1>
<p class="lead">This is a lead paragraph.</p>
<p class="text-muted">This is muted text.</p>
<p class="fw-bold">This is bold text.</p>
```

Exercise 7.2

```
<p class="text-uppercase">this text is uppercase</p>
<p class="text-lowercase">THIS TEXT IS LOWERCASE</p>
<p class="text-capitalize">this text is capitalized</p>
```

8. Forms

Exercise 8.1

```
<form>
  <div class="mb-3">
    <label class="form-label">Full Name</label>
    <input type="text" class="form-control" placeholder="Enter name">
  </div>
  <div class="mb-3">
    <label class="form-label">Email</label>
    <div class="input-group">
      <span class="input-group-text">@</span>
      <input type="email" class="form-control" placeholder="Email">
    </div>
  </div>
  <div class="mb-3 form-check">
    <input type="checkbox" class="form-check-input" id="terms">
    <label class="form-check-label" for="terms">I agree to terms</label>
  </div>
  <button type="submit" class="btn btn-primary">Register</button>
</form>
```

Exercise 8.2

```
<form>
  <div class="form-floating mb-3">
    <input type="email" class="form-control" id="floatEmail"
placeholder="name@example.com">
    <label for="floatEmail">Email address</label>
  </div>
  <div class="form-floating mb-3">
    <input type="password" class="form-control" id="floatPassword" placeholder="Password">
    <label for="floatPassword">Password</label>
  </div>
  <button type="submit" class="btn btn-primary">Login</button>
</form>
```

9. Buttons

Exercise 9.1

```
<button class="btn btn-primary">Primary</button>
<button class="btn btn-secondary">Secondary</button>
<button class="btn btn-success">Success</button>
<button class="btn btn-danger">Danger</button>
<button class="btn btn-warning">Warning</button>
<button class="btn btn-info">Info</button>
<button class="btn btn-light">Light</button>
<button class="btn btn-dark">Dark</button>
```

```
<button class="btn btn-outline-primary">Outline Primary</button>
<button class="btn btn-outline-secondary">Outline Secondary</button>
```

Exercise 9.2

```
<div class="btn-group" role="group">
  <button class="btn btn-primary">Left</button>
  <button class="btn btn-primary">Middle</button>
  <button class="btn btn-primary">Right</button>
</div>

<div class="btn-group mt-3" role="group">
  <input type="checkbox" class="btn-check" id="btncheck1">
  <label class="btn btn-outline-primary" for="btncheck1">Toggle 1</label>

  <input type="checkbox" class="btn-check" id="btncheck2">
  <label class="btn btn-outline-primary" for="btncheck2">Toggle 2</label>
</div>
```

10. Navbars and Navigation

Exercise 10.1

```
<nav class="navbar navbar-expand-lg navbar-light bg-light">
  <div class="container-fluid">
    <a class="navbar-brand" href="#">Logo</a>
    <button class="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-
target="#navMenu">
      <span class="navbar-toggler-icon"></span>
    </button>
    <div class="collapse navbar-collapse" id="navMenu">
      <ul class="navbar-nav me-auto">
        <li class="nav-item"><a class="nav-link" href="#">Home</a></li>
        <li class="nav-item"><a class="nav-link" href="#">About</a></li>
        <li class="nav-item"><a class="nav-link" href="#">Contact</a></li>
      </ul>
      <form class="d-flex">
        <input class="form-control me-2" type="search" placeholder="Search">
        <button class="btn btn-outline-success" type="submit">Search</button>
      </form>
    </div>
  </div>
</nav>
```

Exercise 10.2

```
<ul class="nav nav-tabs">
  <li class="nav-item"><a class="nav-link active" href="#">Home</a></li>
  <li class="nav-item"><a class="nav-link" href="#">Profile</a></li>
  <li class="nav-item"><a class="nav-link" href="#">Contact</a></li>
```

```

</ul>

<ul class="nav nav-pills mt-3">
  <li class="nav-item"><a class="nav-link active" href="#">Home</a></li>
  <li class="nav-item"><a class="nav-link" href="#">Profile</a></li>
  <li class="nav-item"><a class="nav-link" href="#">Contact</a></li>
</ul>

```

11. Cards and Media Objects

Exercise 11.1

```

<div class="card" style="width: 18rem;">
  
  <div class="card-body">
    <h5 class="card-title">John Doe</h5>
    <p class="card-text">Software Developer</p>
    <a href="#" class="btn btn-primary">View Profile</a>
  </div>
</div>

```

Exercise 11.2

```

<div class="d-flex">
  
  <div>
    <h5>Media Heading</h5>
    <p>This is some sample content for the media object layout.</p>
  </div>
</div>

```

12. Spacing Utilities

Exercise 12.1

```

<div class="m-3 p-2 bg-light">Section with margin and padding</div>
<div class="mt-4 py-5 bg-secondary text-white">Section with top margin and vertical padding</div>

```

Exercise 12.2

```

<div class="container py-5">
  <div class="row g-4 text-center">
    <div class="col-md-4">
      <div class="p-4 border rounded">
        <h4 class="mb-3">Basic</h4>
        <p class="mb-4">$9/month</p>
        <button class="btn btn-outline-primary">Choose</button>
      </div>
    </div>
  </div>
</div>

```

```

<div class="p-4 border rounded">
  <h4 class="mb-3">Standard</h4>
  <p class="mb-4">$19/month</p>
  <button class="btn btn-outline-primary">Choose</button>
</div>
</div>
<div class="col-md-4">
  <div class="p-4 border rounded">
    <h4 class="mb-3">Premium</h4>
    <p class="mb-4">$29/month</p>
    <button class="btn btn-outline-primary">Choose</button>
  </div>
</div>
</div>
</div>

```

13. Colors and Backgrounds

Exercise 13.1

```

<div class="p-3 bg-primary text-white">Primary Section</div>
<div class="p-3 bg-warning text-dark">Warning Section</div>
<div class="p-3 bg-success text-white">Success Section</div>
<div class="p-3 bg-danger text-white">Danger Section</div>

```

Exercise 13.2

```

<div class="p-5 bg-gradient bg-dark text-white">
  Gradient Dark Section
</div>

```

14. Display and Visibility

Exercise 14.1

```

<div class="d-none d-md-block">Visible from md and up</div>
<div class="d-none d-lg-flex">Visible as flex from lg and up</div>

```

Exercise 14.2

```

<div class="d-none d-md-block col-md-3 bg-light">
  Sidebar visible on tablets and above
</div>

```

15. Borders, Shadows, and Rounded Corners

Exercise 15.1

```



```

Exercise 15.2

```
<div class="card shadow-lg rounded-pill p-3">
  Card with shadow and pill radius
</div>
```

16. Positioning Utilities

Exercise 16.1

```
<footer class="position-fixed bottom-0 start-0 w-100 bg-dark text-white text-center p-2">
  Fixed Footer Content
</footer>
```

Exercise 16.2

```
<div class="position-relative" style="width:150px;">
  
  <span class="position-absolute top-0 end-0 badge bg-danger">New</span>
</div>
```

17. Icons with Bootstrap Icons

Exercise 17.1

```
<link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/bootstrap-icons@1.11.3/font/bootstrap-
icons.css">

<footer class="text-center p-3">
  <a href="#" class="text-white me-3"><i class="bi bi-facebook"></i></a>
  <a href="#" class="text-white me-3"><i class="bi bi-twitter"></i></a>
  <a href="#" class="text-white me-3"><i class="bi bi-instagram"></i></a>
</footer>
```

Exercise 17.2

```
<button class="btn btn-outline-primary"><i class="bi bi-download"></i></button>
<button class="btn btn-outline-danger"><i class="bi bi-trash"></i></button>
<button class="btn btn-outline-success"><i class="bi bi-check-lg"></i></button>
```

18. Bootstrap 5 JavaScript Plugins

Exercise 18.1

```
<button class="btn btn-primary" data-bs-toggle="modal" data-bs-target="#myModal">
  Open Modal
</button>

<div class="modal fade" id="myModal" tabindex="-1">
  <div class="modal-dialog">
    <div class="modal-content">
      <div class="modal-header">
        <h5 class="modal-title">Modal Title</h5>
        <button type="button" class="btn-close" data-bs-dismiss="modal"></button>
```



```
</div>
<div class="modal-body">
  Modal body content.
</div>
</div>
</div>
</div>
```

Exercise 18.2

```
<div class="accordion" id="accordionExample">
  <div class="accordion-item">
    <h2 class="accordion-header">
      <button class="accordion-button" type="button" data-bs-toggle="collapse" data-bs-
target="#collapseOne">
        Accordion Item 1
      </button>
    </h2>
    <div id="collapseOne" class="accordion-collapse collapse show" data-bs-
parent="#accordionExample">
      <div class="accordion-body">Content for item 1.</div>
    </div>
  </div>
  <div class="accordion-item">
    <h2 class="accordion-header">
      <button class="accordion-button collapsed" type="button" data-bs-toggle="collapse" data-
bs-target="#collapseTwo">
        Accordion Item 2
      </button>
    </h2>
    <div id="collapseTwo" class="accordion-collapse collapse" data-bs-
parent="#accordionExample">
      <div class="accordion-body">Content for item 2.</div>
    </div>
  </div>
</div>
```

19. Customization with Sass

Exercise 19.1

```
npm install bootstrap sass --save-dev
// styles.scss
@import "../node_modules/bootstrap/scss/bootstrap";
npx sass styles.scss styles.css
```

Exercise 19.2

```
// _variables.scss (custom overrides before bootstrap import)
```

```
$primary: #6f42c1;  
$border-radius: 0.75rem;
```

```
// styles.scss  
@import "variables";  
@import "../node_modules/bootstrap/scss/bootstrap";
```